

Target Hamiltonian

ω_q, α

ω_r, g, κ

Physics

Physics features

$\alpha \approx -E_C$

$\omega_q(E_J, E_C)$

Scale and encode

Min-max scaling

Categorical masks

ML

Inverse MLPs

TransmonCross

Ncap Coupler

Resonator

Iterate

Validation

Compare

Percent error

Iterate if needed

Recovered Hamiltonian

$\hat{\omega}_q, \hat{\alpha}$

$\hat{\omega}_r, \hat{g}$

Forward check

Quantum Metal layout

Ansys or surrogate

Decode predictions

Physical units

Valid design fields